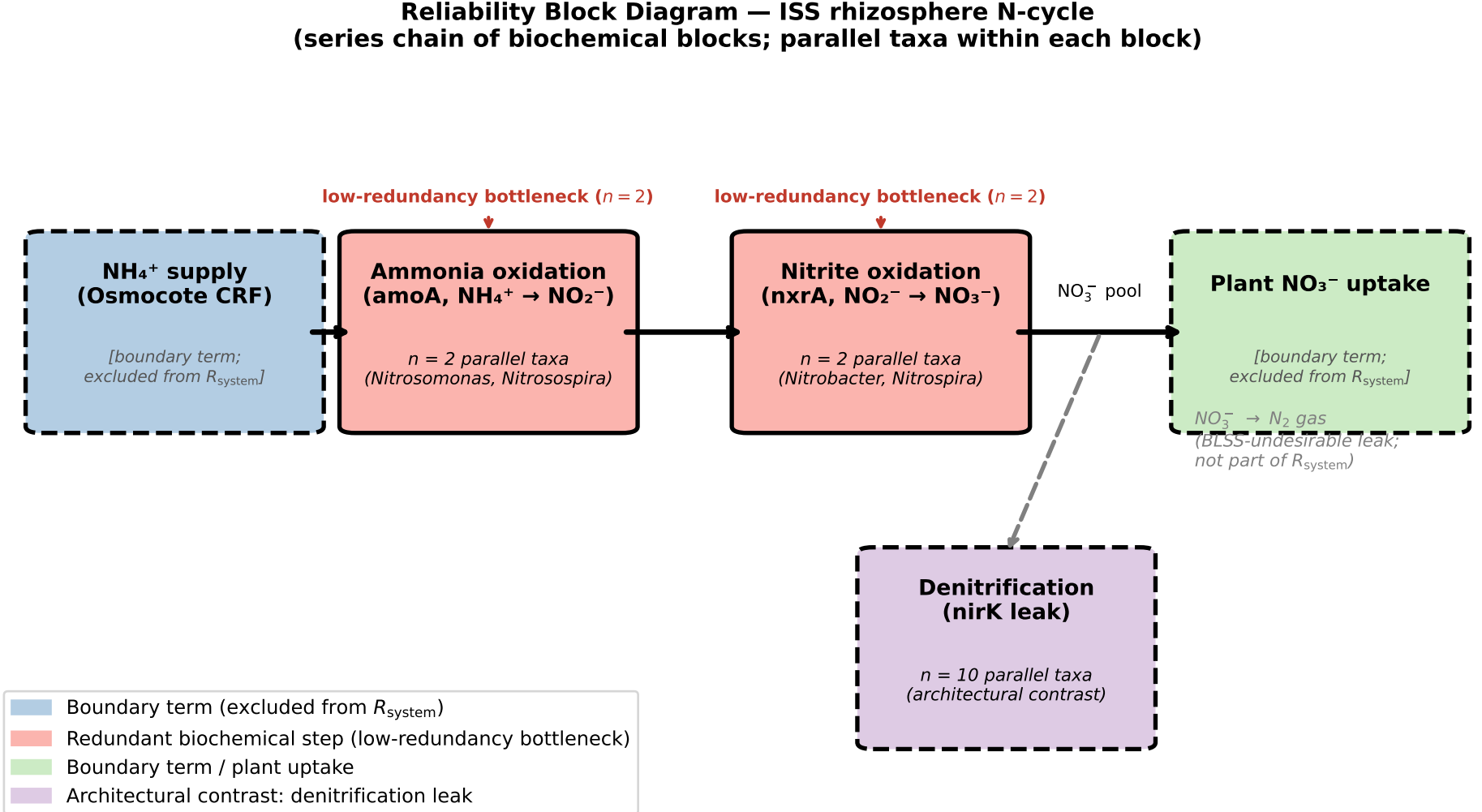
Reliability Block Diagram — ISS rhizosphere N-cycle
(series chain of biochemical blocks; parallel taxa within each block)
low-redundancy bottleneck (n = 2)
low-redundancy bottleneck (n = 2)
NH₄⁺ supply
(Osmocote CRF)
[boundary term;
excluded from R_system]
Ammonia oxidation
(amoA, NH₄⁺ → NO₂⁻)
n = 2 parallel taxa
(Nitrosomonas, Nitrosospira)
Nitrite oxidation
(nxrA, NO₂⁻ → NO₃⁻)
n = 2 parallel taxa
(Nitrobacter, Nitrospira)
NO3− pool
Plant NO₃⁻ uptake
[boundary term;
excluded from R_system]
NO3− → N2 gas
(BLSS-undesirable leak;
not part of R_system)
Denitrification
(nirK leak)
n = 10 parallel taxa
(architectural contrast)
Boundary term (excluded from R_system)
Redundant biochemical step (low-redundancy bottleneck)
Boundary term / plant uptake
Architectural contrast: denitrification leak